Pratyush Potu

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Research Interests

I am generally interested in numerical analysis, and more precisely in applications of topology, geometry, and combinatorics in the analysis of numerical methods for partial differential equations.

Education

Brown University *May 2027*
Ph. D. in Applied Mathematics

- **Thesis Advisor:** Johnny Guzmán

Brown University *May 2025*
M. Sc. in Applied Mathematics

University of Texas at Austin *May 2022*
B. Sc. in Mathematics

- Certificate in Scientific Computation and Data Science

Publications and Preprints

1. Guzmán, J., and Potu, P. A framework for analysis of DEC approximations to Hodge-Laplacian problems using generalized Whitney forms. *Math. Comp.* (2026). https://doi.org/10.1090/mcom/4245
2. Ern, A., Guzmán, J., Potu, P., and Vohralík, M. Local L^2-bounded commuting projections using discrete local problems on Alfeld splits. *Found. Comput. Math.* (2026). to appear, https://arxiv.org/abs/2502.03601
3. Ern, A., Guzmán, J., Potu, P., and Vohralík, M. Discrete Poincaré inequalities: a review on proofs, equivalent formulations, and behavior of constants. *IMA Journal of Numerical Analysis* (2026). https://doi.org/10.1093/imanum/draf089
4. Guzmán, J., Hirani, A. N., Liu, B., and Potu, P. Discrete Poincaré and Bogovskiĭ operators on cochains and Whitney forms. arXiv preprint., https://arxiv.org/abs/2603.29018, 2026
5. Ern, A., Guzmán, J., and Potu, P. Bounded, commuting, discrete-trace preserving projections. arXiv preprint., https://arxiv.org/abs/2604.28103, 2026
6. Potu, P. A symmetric counterexample to Strang's conjecture for bivariate C^1 cubic splines on triangulations. arXiv preprint., https://arxiv.org/abs/2610.08470, 2026
7. Guzmán, J., Potu, P., and Tyssvang, K. O. Discrete Poincaré and Bogovskiĭ operators for higher order finite element de Rham complexes. arXiv preprint., https://arxiv.org/abs/2611.05698, 2026
8. Izmailov, A., Keith, B., and Potu, P. Proximal Galerkin for total bounded variation based image denoising. Work in progress., 2026

Selected Presentations

- Finite Element Circus, University of Connecticut, Avery Point, Groton, CT, Oct. 2026
- Workshop on Discretization of Differential Complexes, Erwin Schrödinger Institute, Vienna, Austria, May 2026
- Finite Element Circus, Tufts University, Medford, MA, Apr. 2026
- AMS Spring Eastern Sectional Meeting, Boston College, Chestnut Hill, MA, Mar. 2026
- Finite Element Circus, George Mason University, Arlington, VA, Oct. 2025
- IMSI Workshop on Discrete Exterior Calculus (Poster), Chicago, IL, Sep. 2025
- EMS School on Mathematical Modelling, Numerical Analysis and Scientific Computing, Kácov, Czech Republic, Jun. 2025

- AMS Spring Eastern Sectional Meeting, Hartford, CT, Apr. 2025
- Finite Element Circus, Oakland University, Rochester, MI, Apr. 2025
- SIAM Conference on Computational Science and Engineering (CSE25), Fort Worth, TX, Mar. 2025

Conference and Workshop Attendance

- Workshop on Simulation-Based Optimization with Applications, ICERM, Providence, RI, Apr. 2026
- Semester Program in Numerical PDEs, ICERM, Providence, RI, Jan.-May 2024
- Finite Element Circus, Brown University, Providence, RI, Apr. 2024
- Finite Element Circus, Bridgewater State University, Bridgewater, MA, Mar. 2023

Teaching and Mentoring

Graduate Teaching Assistant *Sep 2023 - May 2024*
Brown University

- Statistical Inference I (Fall 2023)
- Applied Partial Differential Equations I (Spring 2024)

Directed Reading Program *Fall 2023 - Fall 2025*
Brown University

- An extracurricular activity where graduate students mentor undergraduate students in a reading project over the course of a semester. Graduate students are responsible for planning the project, weekly meetings and assistance producing a poster.
- Topics mentored: Calculus of Variations (one student), Numerical Linear Algebra (one student), Spectral Graph Theory (three students), Splines (one student), Numerical Analysis (one student).

Academic Service

- Brown Society for Industral and Applied Mathematics (SIAM) Student Chapter (Secretary Sep. 2023 - May 2024, President Sep. 2024 - May 2026, Vice President Sep. 2026 - May 2027)
- Brown Applied Math Graduate Student Seminar Organizer (Sep. 2023 - May 2027)
- Reviewer for IMA Journal of Numerical Analysis and Calcolo

Skills

Software: Python (FEniCS, Firedrake, Pytorch), C++ (MFEM)

Languages: English, Telugu